

Leading The Charge

War is hell, and people die. However, technology can help minimise losses, and Shadab Ansari found a way...

Samir Makwana

Seven years ago, many brave Indian soldiers sacrificed themselves at Kargil to protect our freedom. Our warriors, who stand along our nation's borders risking life and limb need all the help they can get. It took the ingenuity of a 23-year-old from Lucknow to figure out a way in which tech could save lives on the front line!

Our borders are some of the most treacherous places on earth, and it's hard to spot enemy encampments or track enemy movement; still, soldiers risk their lives trying to spy on the enemy. But what if you didn't need soldiers to do this? A robot could provide accurate reconnaissance without endangering lives...

That's exactly what Shadab Ahmad Ansari thought of when building his invention, *Soldier Gear*.

Soldier Gear is an espionage robot controlled using a special military suit. It has a camera and microphone on it, which is used to send information about any location. This gear is to

be used in defence warfare when vital information of enemy troops' location and strength are sought.

How much did it cost?

"Though the project was made out of components bought from the scrap market, it still cost me Rs 10,000. Most of the expenses were due to the trial-and-error method I had to use when making the circuits. If I had better knowledge of robotics and circuits, it would have worked out much cheaper," says Shadab.

What made him want to do it?

Shadab reminisces, "I was watching a movie based on the Kargil war. The miserable situation of Indian soldiers depicted in the movie made me want to create something to help them. Since I had already assisted a senior with his project of making a simple robot, I came up with a vague idea of making an advanced robot. I started from scratch-researching robotics on the Net."

Building Blocks

Soldier Gear comprises a printed circuit board, gearboxes of cars, parts of remote controlled toy cars, rotors, relays, batteries of various voltages, a machine belt, a vacuum cleaner hood, rubber bands, and various other everyday items bought from the local scrap market. Shadab actually used commonly-used adhesives as well as rubber bands to hold the components together!

